Assignment - 01

1) Design a deterministic finite automata (DFA) to accept strings that start and end with same symbols over $\Sigma = \{0,1\}$. Write the formal definition of the DFA. Also show that the string 010100 is accepted by the DFA.

To construct a deterministic finite automation (DFA) that accepts the binary strings whose first and last symbols are the same.

Formal definition:-

A DFA is defined as a five tuple:

$$M = (Q, \Sigma, \delta, q_0, F).$$

where:

Q = set of states

Σ = input alphabet

δ = Transition function

q_0 = Initial State

F = Set of final States.

1. Set of State:

$$Q = \{q_0, q_1, q_2, q_3, q_4\}.$$

2. Input alphabet:

$$\Sigma = \{0,1\}$$

3. Transition Table

current State	Input 0	Input 1
$\rightarrow q_0$	q_1	q_3
$*q_1$	q_1	q_2
q_2	q_1	q_2
$*q_3$	q_4	q_3
q_4	q_4	q_3

4. Initial State

q_0

5. Final States

6. verification of the string 010100

Given string

$\omega = 010100$

process each symbol using the transition table

step	Input symbol	current state.
0	–	q_0
1	0	q_1
2	1	q_2
3	0	q_1
4	1	q_2
5	0	q_1
6	0	q_1

Therefore,

$$q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \xrightarrow{0} q_1 \xrightarrow{0} q_1$$

Result:- The string 010100 is accepted by the DFA.

2) Design NFA to accept string having 101 as a substring.

To construct a nondeterministic finite automation (NFA) that accepts all binary strings containing 101 as a substring.

1). Set of states

$Q = \{q_0, q_1, q_2, q_3\}$.

2). Input alphabet

$\Sigma = \{0,1\}$.

3) Transition table

current state	Input 0	Input 1.
$\rightarrow q_0$	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$
$*q_3$	$\{q_3\}$	$\{q_3\}$.

Here $\emptyset$ represents an empty set of transitions.

4). final state

$F = \{q_3\}$.

Example:

Consider the string $w = 11010$

The NFA can follow this path:

$$q_0 \xrightarrow{1} q_1 \xrightarrow{1} \text{(this path stops)}$$

$$q_0 \xrightarrow{1} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_2$$

The path does not accept, because the input contains 101 starting at the second symbol, so the correct accepting path is.

$$q_0 \xrightarrow{1} q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_3 \xrightarrow{0} q_3$$

Result; The NFA accepts all binary strings containing 101 as a substring.

3) Apply the principle of mathematic induction and prove that for all positive integers n, the statement given below is true.

$$\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$$

proof by mathematical induction.

Step 1: Base case

For $n=1$, the left hand side (LHS) is

$$LHS = \sum_{i=1}^{1} i(i+1)$$

$$= 1(1+1)$$

$$= 1(2) = 2.$$

The right hand side (RHS) is

$$RHS = \frac{n(n+1)(n+2)}{3}$$

substituting $n=1$,

$$= \frac{1(1+1)(1+2)}{3}$$

$$= \frac{1 \times 2 \times 3}{3} = 2$$

Therefore,

$$LHS = RHS = 2.$$

Step 2: Induction Hypothesis

Assume, $n=k$

That is $\sum_{i=1}^{k} i(i+1) = \frac{k(k+1)(k+2)}{3}$

step3 : Induction Step.

we must prove that n=k+1

LHS: $\sum_{i=1}^{k+1} i(i+1)$

$= \sum_{i=1}^{k} i(i+1) + (k+1)(k+2)$

using the induction hypothesis:

$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$.

$= (k+1)(k+2)\left(\frac{k}{3}+1\right)$

$= (k+1)(k+2)\left(\frac{k+3}{3}\right)$

$= \frac{(k+1)(k+2)(k+3)}{3}$.

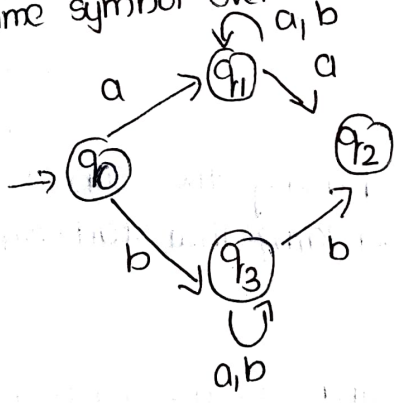
R.H.S $\Rightarrow n = k+1$

$\frac{n+((k+1)(k+1)+1)((k+1)+2)}{3}$

Therefore:

$$\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$$

4) consider the NFA given below which accepts the language representing the set of all strings that start and end with the same symbol over the alphabet $\Sigma = \{a, b\}$.



To construct a deterministic finite automation (DFA) equivalent to the given NFA which accepts strings that start and end with the same symbol over

$\Sigma = \{a, b\}$.

1. NFA transition table:

State	Input a	Input b
$\rightarrow q_0$	$\{q_1\}$	$\{q_3\}$
q_1	$\{q_1, q_2\}$	$\{q_1\}$
$*q_2$	$\emptyset$	$\emptyset$
q_3	$\{q_3\}$	$\{q_2, q_3\}$

2. Apply the subset construction method

let the DFA states be:

$A = \{q_0\}$

$B = \{q_1\}$

$C = \{q_3\}$

$D = \{q_1, q_2\}$

$E = \{q_3, q_2\}$.

$F = \{D, E\}$.

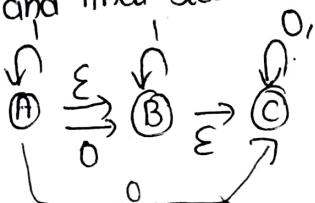
3. DFA Transition table

DFA State	Input a	Input b
$\rightarrow A$	B	C
B	D	B
C	C	E
* D	D	B
* E	C	E

Result:-

The equivalent DFA is constructed using the subset construction method. It accepts strings over $\{a, b\}$ that start and end with the same symbol.

5. A NFA with ϵ-transitions is given below. Find ϵ-closure for each state and construct an equivalent NFA without ϵ-transitions that accepts the same language. Draw its transition table and mark the initial and final states.



To find the ε-closure of each state and construct an equivalent NFA without ε-transitions.

Step 1: Find ε-closure.

- ε-closure (A) = {A, B, C}
- ε-closure (B) = {B, C}
- ε-closure (C) = {C}.

Step 2: construct the Transition Table.

State	Input 0	Input 1
→*A	{B, C}	{A, B, C}
*B	{C}	{B, C}
*C	{C}	{C}.

Step 3: Initial and final States.

Initial State:

$q_0 = A$

Final States:

$F = \{A, B, C\}$.

Result:-

The ε-closure of each state was calculated and the equivalent NFA without ε-transitions was constructed successfully.